B. Collatz Conjecture

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1200 | Tags: brute force, implementation, math, number theory | https://codeforces.com/problemset/problem/1982/B

Recently, the first-year student Maxim learned about the Collatz conjecture, but he didn't pay much attention during the lecture, so he believes that the following process is mentioned in the conjecture:

There is a variable x and a constant y. The following operation is performed k times:

- increase x by 1, then
- while the number x is divisible by y, divide it by y.

Note that both of these actions are performed sequentially within one operation.

For example, if the number $x = 16$, $y = 3$, and $k = 2$, then after one operation x becomes 17, and after another operation x becomes 2, because after adding one, $x = 18$ is divisible by 3 twice.

Given the initial values of x, y, and k, Maxim wants to know what is the final value of x.

Input

Each test consists of multiple test cases. The first line contains an integer t ($1 \le t \le 10^4$) — the number of test cases. Then follows the description of the test cases.

The only line of each test case contains three integers x, y, and k ($1 \le x, k \le 10^9$, $2 \le y \le 10^9$) — the initial variable, constant and the number of operations.

Output

For each test case, output a single integer — the number obtained after applying k operations.

Example

Input
13 1 3 1 2 3 1 24 5 5 16 3 2 2 2 1 1337 18 1 1 2 144133 12345678 3 10 998244353 2 998244353 998244353 123456789 998244352 998244354 998241111 998244352 998244355 2 9982443 1000000000 1000000000 1000000000

Output

```
2
1
1
2
3
1338
1
16936
1
21180097
6486
1
2
```

Note

In the first test case, there is only one operation applied to $x = 1$, resulting in x becoming 2.

In the second test case, for $x = 2$, within one operation, one is added to x and it's divided by $y = 3$, resulting in x becoming 1.

In the third test case, x changes as follows:

- After the first operation, $x = 1$, because $24 + 1 = 25$ and 25 is divisible by $y = 5$ twice within one operation.
- After the second operation, $x = 2$.
- After the third operation, $x = 3$.
- After the fourth operation, $x = 4$.
- After the fifth operation, $x = 1$.